

TrES-3b

R-2500

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_110522 · created 2026-08-24T23:12:31+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2455703.7012 +/- 0.00092 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.186 +/- 0.037
Transit depth (Rp/Rs)^2	3.47 +/- 1.37 [%]
Orbital Inclination (inc)	81.16 +/- 1.22
Airmass coefficient 1 (a1)	0.984 +/- 0.014
Airmass coefficient 2 (a2)	0.0074 +/- 0.0038
Scatter in the residuals of the lightcurve fit is	1.4 %
Best Comparison Star	#3 - [504, 102]
Optimal Aperture	3.47
Optimal Annulus	6.59
Transit Duration (day)	0.047 +/- 0.017

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.186 ± 0.037 vs 0.16309 (deviation 0.51σ)
Duration fit vs published	0.047 d vs 0.0567 d (deviation 0.47σ)
Mid-time O–C	1.6 min
Depth SNR	4.1
Residual scatter	1.4 %

Light curve

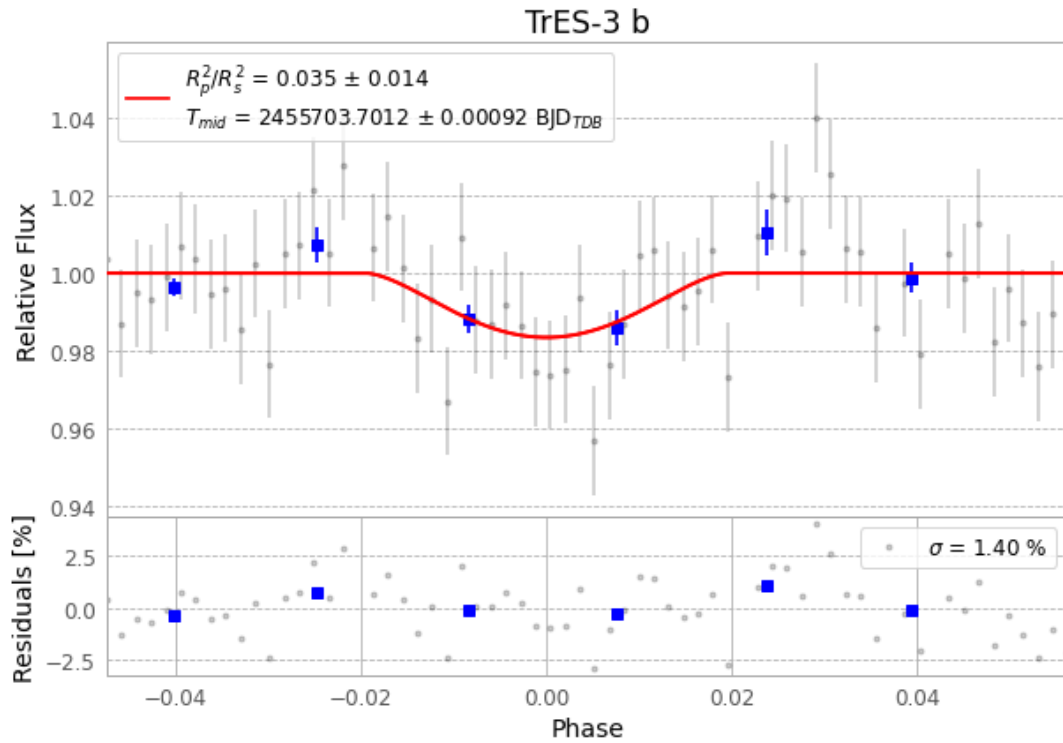


Figure 1 — FinalLightCurve_TrES-3 b_2011-05-21.png (SHA-256 09319693d55065cf...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [294, 206], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 4437b70af625b5f3...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

66 FITS frames (43 MB), TRES-3110522031512.FITS → TRES-3110522063013.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-110522.log (b5043a3ab656...), FinalLightCurve_TrES-3 b_2011-05-21.png (09319693d550...), FinalLightCurve_TrES-3 b_2011-05-21.csv (c1071eff940f...)

Compute resources

Wall time 112.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-24 23:12Z.